

The Mathemaniac

IIT Foundation & Olympiad Module for Class IX

Chemistry

Theme 1: Language of Chemistry

1. Symbols, valencies, radicals and chemical formulae
2. Writing and balancing chemical equations
3. Relative atomic mass and molecular mass

Theme 2: Chemical Changes and Reactions

1. Physical and chemical changes
2. Exothermic and endothermic reactions
3. Indicators of chemical reactions

Theme 3: Water

1. Water as a universal solvent
2. Hard and soft water
3. Purification and treatment of water

Theme 4: Atomic Structure and Bonding

1. Atomic structure, isotopes and electronic configuration
2. Octet rule
3. Electrovalent and covalent bonding

Theme 5: Periodic Table

1. Historical development of periodic classification
2. Modern periodic table, groups and periods
3. Periodic trends

Theme 6: Hydrogen

1. Position and properties of hydrogen
2. Preparation, collection and uses
3. Oxidation and reduction

Theme 7: Olympiad Enrichment

1. Mole concept basics
2. Advanced equation balancing
3. Periodic trend reasoning
4. Higher-order conceptual MCQs